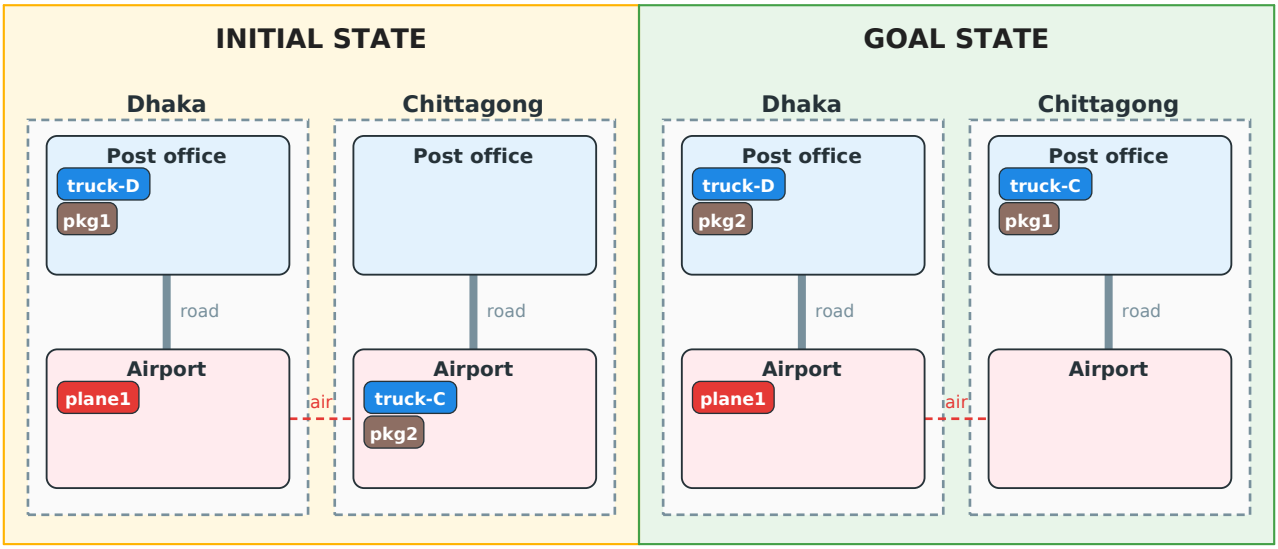
Problem 8: Logistics

Level: **Intermediate / Medium** · Folder: **08_Logistics/** · Files: domain.pddl, problem.pddl, solution_plan.txt

1. The problem

There are two cities, **Dhaka** and **Chittagong**, and each has a post office and an airport. A truck works inside each city, and one airplane flies between the airports. Deliver **pkg1** from the Dhaka post office to the Chittagong post office, and **pkg2** from Chittagong airport to the Dhaka post office.



The goal picture shows one state that satisfies the goal (the final state of the plan). Only the facts in the :goal section are required; anything else may be different.

What you will learn

- A deeper **type hierarchy** (truck and airplane are vehicles, airport is a location, and so on).
- **Generic actions**: one load/unload works for both trucks and airplanes.
- Restricting movement with static facts (*in-city*) and with types (fly only between airports).
- Coordinating several vehicles in a longer plan (15 steps).

2. Formulating the problem in PDDL

Step 1: Types and objects. city: dhaka, chittagong. location: dhaka-po, ctg-po. airport: dhaka-airport, ctg-airport. truck: truck-dhaka, truck-ctg. airplane: plane1. package: pkg1, pkg2.

Step 2: Predicates describe what can be true or false in a state:

Predicate	Meaning
`(at ?x ?l)`	Package or vehicle ?x is at location ?l (?x has type *locatable*).
`(in ?p ?v)`	Package ?p is inside vehicle ?v.
`(in-city ?l ?c)`	Location ?l belongs to city ?c (static).

Step 3: Actions. Each action has *parameters*, a *precondition* (what must be true to apply it) and an *effect* (facts that become **true (+)** or **false (−)**):

Action	Precondition	Effect
load `?p - package, ?v - vehicle, ?l - location` Put package ?p into vehicle ?v when both are at ?l.	`(at ?p ?l)` `(at ?v ?l)`	`+ (in ?p ?v)` `− (at ?p ?l)`
unload `?p - package, ?v - vehicle, ?l - location` Take package ?p out of vehicle ?v at ?l.	`(in ?p ?v)` `(at ?v ?l)`	`+ (at ?p ?l)` `− (in ?p ?v)`

drive-truck
`?t - truck, ?from - location, ?to - location, ?c - city`
Drive truck ?t between two locations of the same city ?c.
	`(at ?t ?from)`
	`(in-city ?from ?c)`
	`(in-city ?to ?c)`
		`+ (at ?t ?to)`
		`− (at ?t ?from)`

fly-airplane `?a - airplane, ?from - airport, ?to - airport` Fly airplane ?a between any two airports.	`(at ?a ?from)`	`+ (at ?a ?to)` `− (at ?a ?from)`

Step 4: Initial state (`:init`) lists *every* fact that is true at the start. Everything else is false (closed-world assumption). **Step 5: Goal** (`:goal`) lists the facts that must be true at the end.

INITIAL STATE (9 facts)	GOAL

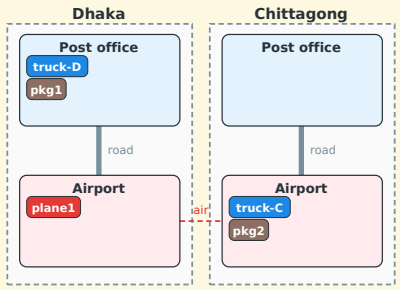
`(in-city dhaka-po dhaka)`
`(in-city dhaka-airport dhaka)`
`(in-city ctg-po chittagong)`
`(in-city ctg-airport chittagong)`
`(at truck-dhaka dhaka-po)`
`(at truck-ctg ctg-airport)`
`(at plane1 dhaka-airport)`
`(at pkg1 dhaka-po)`
`(at pkg2 ctg-airport)`
	`(at pkg1 ctg-po)`
	`(at pkg2 dhaka-po)`

Modelling notes

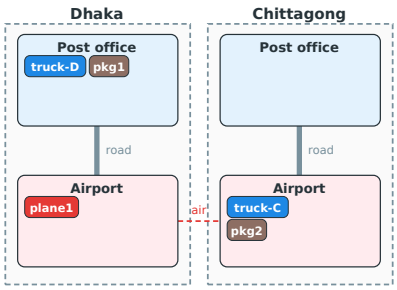
- Type tree: `truck, airplane < vehicle < locatable`; `package < locatable`; `airport < location`. Because an airport *is a* location, trucks can drive to it and packages can wait there.
- drive-truck has a parameter **?c** that is used only in the precondition. It forces both locations to be in the **same** city.
- There is no direct truck-to-plane transfer. A package is unloaded at the airport and then loaded again.
- The plane carries pkg1 to Chittagong, picks up pkg2 there and brings it back. No trip is wasted.

3. Solution plan (15 actions, optimal)

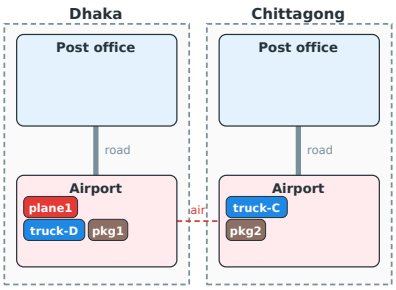
The planner uses the plane for both directions and both trucks for the local legs. (In the pictures, truck-D = truck-dhaka and truck-C = truck-ctg; a package badge next to a vehicle is *inside* it.) The pictures below show the state **after** each action. They were produced by executing the plan on the PDDL model.



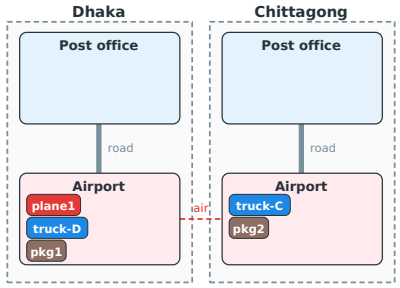
Initial state



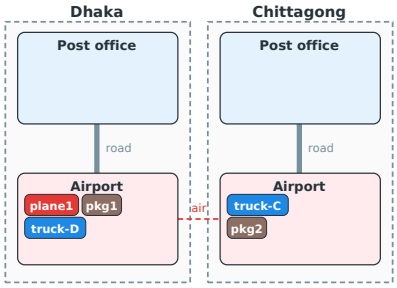
Step 1
(load pkg1 truck-dhaka dhaka-po)



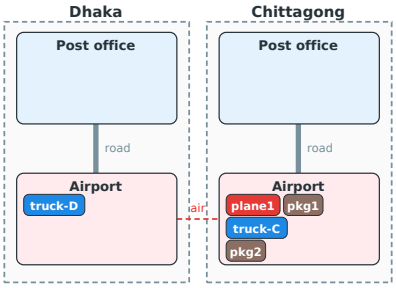
Step 2
(drive-truck truck-dhaka dhaka-po dhaka-airport dhaka)



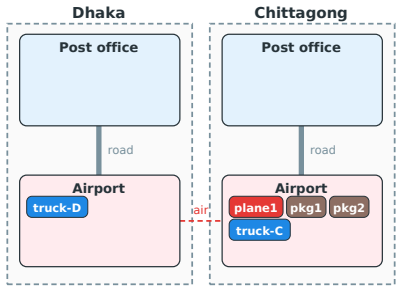
Step 3
(unload pkg1 truck-dhaka dhaka-airport)



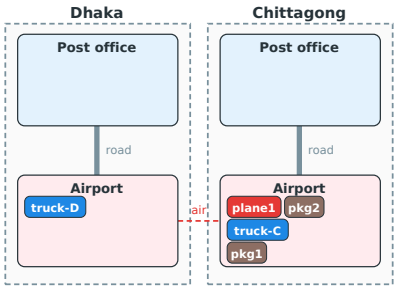
Step 4
(load pkg1 plane1 dhaka-airport)



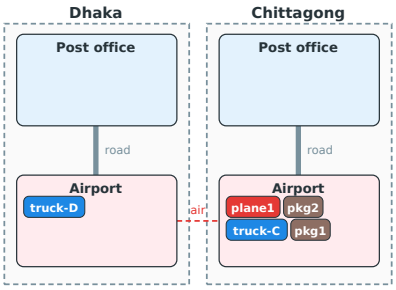
Step 5
(fly-airplane plane1 dhaka-airport ctg-airport)



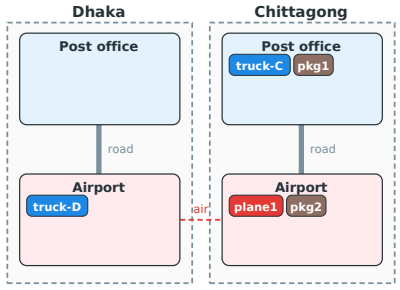
Step 6
(load pkg2 plane1 ctg-airport)



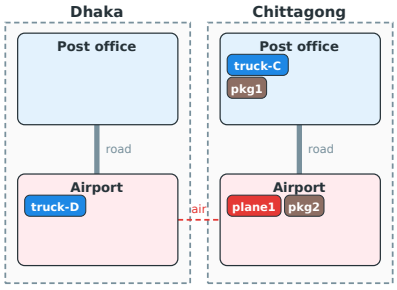
Step 7
(unload pkg1 plane1 ctg-airport)



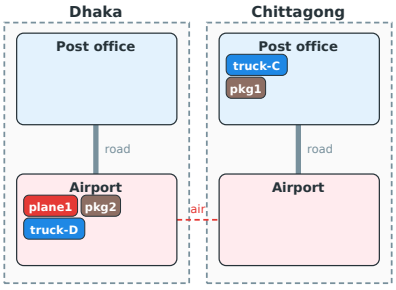
Step 8
(load pkg1 truck-ctg ctg-airport)



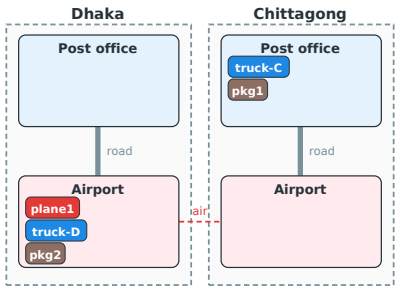
Step 9
(drive-truck truck-ctg ctg-airport ctg-po chittagong)



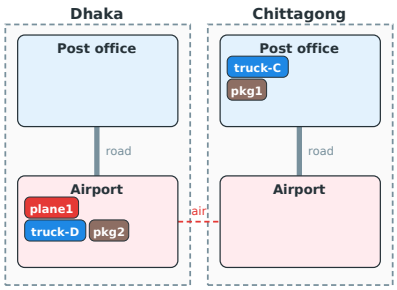
Step 10
(unload pkg1 truck-ctg ctg-po)



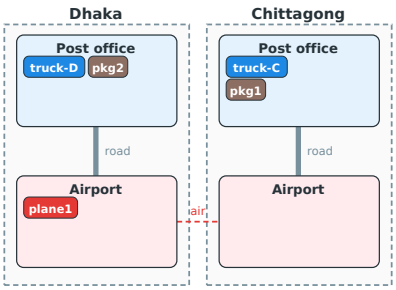
Step 11
(fly-airplane plane1 ctg-airport dhaka-airport)



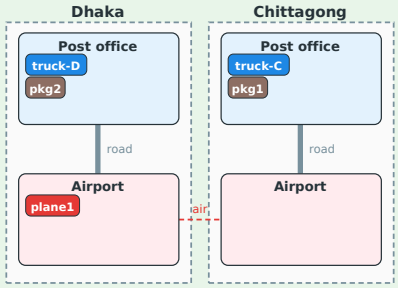
Step 12
(unload pkg2 plane1 dhaka-airport)



Step 13
(load pkg2 truck-dhaka dhaka-airport)



Step 14
(drive-truck truck-dhaka dhaka-airport dhaka-po dhaka)



Step 15
(unload pkg2 truck-dhaka dhaka-po)

4. How to run it

Online (no installation)	Open **editor.planning.domains**, paste *domain.pddl* and *problem.pddl* into two tabs and press **Solve**.
pyperplan (Python)	`pip install pyperplan` `pyperplan domain.pddl problem.pddl` (breadth-first search) `pyperplan -s astar -H lmcut domain.pddl problem.pddl` (optimal A*)
Fast Downward	`fast-downward.py domain.pddl problem.pddl --search "astar(lmcut())"`

5. Lab exercises

1. Add a third city *Sylhet* (post office + airport + truck) and a package going there.
2. Add a package that only moves inside Dhaka. Which vehicles does the plan use?
3. Start the airplane at Chittagong airport. How does the plan change?

6. PDDL source code (appendix)

domain.pddl

```
;; ============================================================
;; Problem 8 : Logistics
;; Level     : Intermediate / Medium
;; Trucks carry packages inside a city; airplanes fly packages
;; between airports of different cities.
;; ============================================================
(define (domain logistics)
  (:requirements :strips :typing)

  (:types
    city location locatable - object
    airport        - location     ; an airport is a special location
    package vehicle - locatable    ; things that can be "at" a location
    truck airplane  - vehicle
  )

  (:predicates
    (at ?x - locatable ?l - location)   ; package/vehicle ?x is at ?l
    (in ?p - package ?v - vehicle)      ; package ?p is inside vehicle ?v
    (in-city ?l - location ?c - city)   ; location ?l belongs to city ?c
  )

  ;; Load a package into any vehicle at the same location
  (:action load
    :parameters (?p - package ?v - vehicle ?l - location)
    :precondition (and (at ?p ?l) (at ?v ?l))
    :effect (and (in ?p ?v) (not (at ?p ?l)))
  )

  ;; Unload a package from a vehicle
  (:action unload
    :parameters (?p - package ?v - vehicle ?l - location)
    :precondition (and (in ?p ?v) (at ?v ?l))
    :effect (and (at ?p ?l) (not (in ?p ?v)))
  )

  ;; A truck drives between two locations of the SAME city
  (:action drive-truck
    :parameters (?t - truck ?from - location ?to - location ?c - city)
    :precondition (and (at ?t ?from)
                       (in-city ?from ?c) (in-city ?to ?c))
    :effect (and (at ?t ?to) (not (at ?t ?from)))
  )

  ;; An airplane flies between two airports (any cities)
  (:action fly-airplane
    :parameters (?a - airplane ?from - airport ?to - airport)
    :precondition (at ?a ?from)
    :effect (and (at ?a ?to) (not (at ?a ?from)))
  )
```

```
)
```

problem.pddl

```
;; Two cities: Dhaka and Chittagong. Each has a post office and an airport.
;;  pkg1 : Dhaka post office      -> Chittagong post office
;;  pkg2 : Chittagong airport     -> Dhaka post office
(define (problem logistics-bd)
  (:domain logistics)

  (:objects
    dhaka chittagong        - city
    dhaka-po ctg-po         - location
    dhaka-airport ctg-airport - airport
    truck-dhaka truck-ctg   - truck
    plane1                  - airplane
    pkg1 pkg2               - package
  )

  (:init
    (in-city dhaka-po dhaka)  (in-city dhaka-airport dhaka)
    (in-city ctg-po chittagong) (in-city ctg-airport chittagong)

    (at truck-dhaka dhaka-po)
    (at truck-ctg ctg-airport)
    (at plane1 dhaka-airport)

    (at pkg1 dhaka-po)
    (at pkg2 ctg-airport)
  )

  (:goal (and (at pkg1 ctg-po)
              (at pkg2 dhaka-po)))
)
```

solution_plan.txt

```
 1. (load pkg1 truck-dhaka dhaka-po)
 2. (drive-truck truck-dhaka dhaka-po dhaka-airport dhaka)
 3. (unload pkg1 truck-dhaka dhaka-airport)
 4. (load pkg1 plane1 dhaka-airport)
 5. (fly-airplane plane1 dhaka-airport ctg-airport)
 6. (load pkg2 plane1 ctg-airport)
 7. (unload pkg1 plane1 ctg-airport)
 8. (load pkg1 truck-ctg ctg-airport)
 9. (drive-truck truck-ctg ctg-airport ctg-po chittagong)
10. (unload pkg1 truck-ctg ctg-po)
11. (fly-airplane plane1 ctg-airport dhaka-airport)
12. (unload pkg2 plane1 dhaka-airport)
13. (load pkg2 truck-dhaka dhaka-airport)
14. (drive-truck truck-dhaka dhaka-airport dhaka-po dhaka)
15. (unload pkg2 truck-dhaka dhaka-po)
```